

CSD101 - Discrete Structures (Discrete Mathematics) Fall 2016

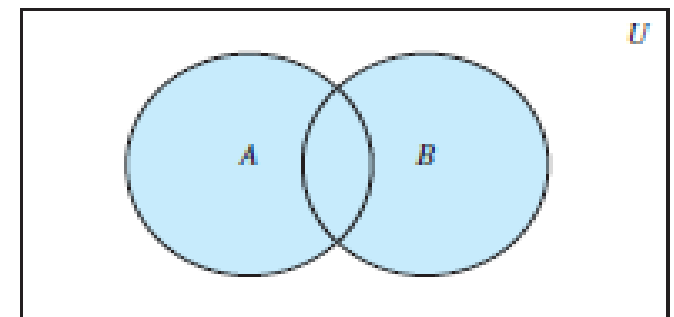
Lecture - 9 Set Operations

Set Operations

- Two sets can be combined in many different ways.
- Set operations can be used to combine sets.

Union

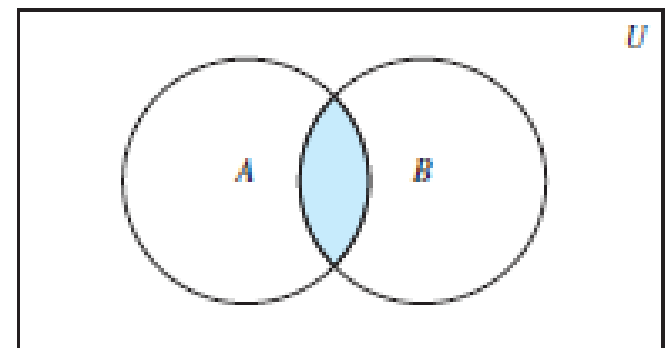
- Let A and B be sets.
- The **union** of A and B , denoted by $A \cup B$, is the set containing those elements that are either in A or in B , or in both.
- $A \cup B = \{x \mid x \in A \vee x \in B\}$



$A \cup B$ is shaded.

Intersection

- Let A and B be sets.
- The **intersection** of A and B , denoted by $\mathbf{A \cap B}$, is the set containing those elements in both A and B .
- $A \cap B = \{x \mid x \in A \wedge x \in B\}$



$A \cap B$ is shaded.

Union (example)

- Let $A = \{1, 2, 3\}$
 $B = \{2, 4, 6, 8\}$
 $A \cup B = \{1, 2, 3, 4, 6, 8\}$
- Let $A = \{x \mid x \in \mathbf{Z} \wedge x \text{ is even}\}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$
 $A \cup B = \mathbf{Z}$

Intersection (example)

- Let $A = \{1, 2, 3\}$
 $B = \{2, 4, 6, 8\}$
 $A \cap B = \{2\}$
- Let $A = \mathbf{Z}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$
 $A \cap B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$

Disjoint Sets

- Two sets are called **disjoint** if their intersection is empty.
- Let $A = \{x \mid x \in \mathbf{Z} \wedge x \text{ is even}\}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$
 $A \cap B = \emptyset$

The Cardinality of the Union of Sets

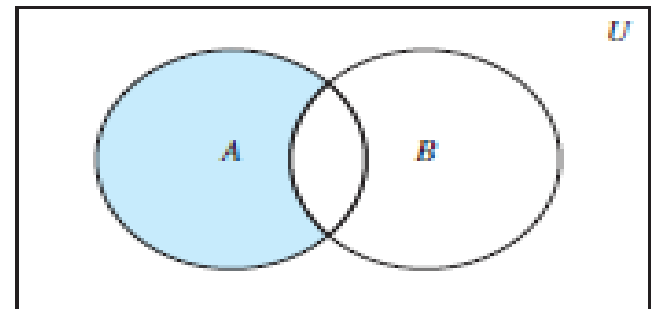
- $|A \cup B| = ?$

Solution:

- Let $A = \{1, 2, 3\}$
 $B = \{2, 3, 4\}$
 $A \cup B = \{1, 2, 3, 4\}$
- $|A| = 3$ $|B| = 3$ $|A \cup B| = 4$
- **$|A \cup B| = |A| + |B| - |A \cap B|$**

Difference

- Let A and B be sets.
- The **difference** of A and B, denoted by **$A - B$** , is the set containing those elements that are in A but not in B. (also called **complement of B with respect to A**).
- $A - B = \{x \mid x \in A \wedge x \notin B\}$



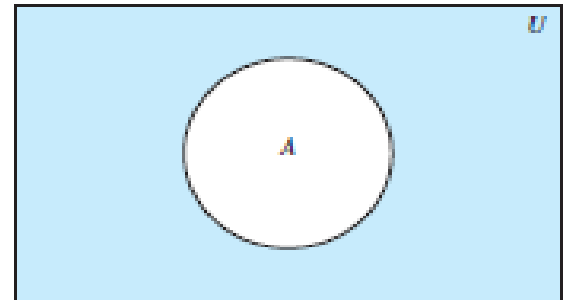
$A - B$ is shaded.

Difference (example)

- Let $A = \{1, 2, 3\}$
 $B = \{2, 4\}$
 $A - B = \{1, 3\}$
- Let $A = \mathbf{Z}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd} \}$
 $A - B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is even} \}$

Complement

- Let U be the universal set and A be a set.
- The **complement** of A , denoted by $\bar{A}$, is the complement of A with respect to U (which is $U - A$).
- $\bar{A} = \{ x \mid x \notin A \}$



$\bar{A}$ is shaded.

Complement (example)

- Let $A = \{ a, b, c, d \}$ and
U is the set of English alphabet
 $\bar{A} = \{ e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z \}$
- Let $A = \{ x \mid x \in \mathbf{Z} \wedge x \text{ is odd} \}$ and
U is $\mathbf{Z}$
 $\bar{A} = \{ x \mid x \in \mathbf{Z} \wedge x \text{ is even} \}$

Summary Set Operations

Operation	Notation
Union	$A \cup B = \{x \mid x \in A \vee x \in B\}$
Intersection	$A \cap B = \{x \mid x \in A \wedge x \in B\}$
Difference	$A - B = \{x \mid x \in A \wedge x \notin B\}$
Complement ($U - A$)	$\bar{A} = \{x \mid x \notin A\}$

Set Identities

$$A \cup \emptyset = A$$

$$A \cap U = A$$

Identity Laws

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

Domination Laws

$$A \cup A = A$$

$$A \cap A = A$$

Idempotent Laws

$$\overline{(\overline{A})} = A$$

Complementation Law

$$A \cup \overline{A} = U$$

$$A \cap \overline{A} = \emptyset$$

Complement Laws

Set Identities

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Commutative Laws

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Associative Laws

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Absorption Laws

Set Identities

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

De Morgan's Law

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Distributive Law

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

How to Prove a Set Identity

- Four methods:
 - Use the basic set identities
 - Use membership tables
 - Prove each set is a subset of each other
 - Use set builder notation and logical equivalences

Set Identities (example)

- Show $\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$

Solution:

$$\begin{aligned}\overline{A \cup (B \cap C)} &= \bar{A} \cap (\overline{B \cap C}) && \text{(By DeMorgan's Law)} \\ &= \bar{A} \cap (\bar{B} \cup \bar{C}) && \text{(By DeMorgan's Law)} \\ &= \bar{A} \cap (\bar{C} \cup \bar{B}) && \text{(By Commutative Law)} \\ &= (\bar{C} \cup \bar{B}) \cap \bar{A} && \text{(By Commutative Law)}\end{aligned}$$

What is a membership table

- Membership tables show all the combinations of sets an element can belong to
 - 1 means the element belongs, 0 means it does not
- Consider the following membership table:

A	B	$A \cup B$	$A \cap B$	$A - B$
1	1	1	1	0
1	0	1	0	1
0	1	1	0	0
0	0	0	0	0

Membership Table

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Distributive Law

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Proof by showing each set is a subset of the other

- Assume that an element is a member of one of the identities
 - Then show it is a member of the other

Example

- Show $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Solution:

Part 1: $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$

Assume $x \in A \cap (B \cup C)$

$(x \in A) \wedge (x \in (B \cup C))$ Definition of Intersection

$(x \in A) \wedge (x \in B \vee x \in C)$ Definition of Union

$(x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)$ Distributive Law

$(x \in (A \cap B)) \vee (x \in (A \cap C))$ Definition of Intersection

$x \in (A \cap B) \cup (A \cap C)$ Definition of Union

So, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$

Example

- Show $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Solution:

Part 2: $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

Assume $x \in (A \cap B) \cup (A \cap C)$

$(x \in A \cap B) \vee (x \in A \cap C)$ Definition of Union

$(x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)$ Definition of Intersection

$x \in A \wedge (x \in B \vee x \in C)$ Distributive Law

$x \in A \wedge (x \in B \cup C)$ Definition of Union

$x \in A \cap (B \cup C)$ Definition of Intersection

So, $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

Thus, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof by set builder notation and logical equivalences

- First, translate both sides of the set identity into set builder notation
- Then use one side (or both) to make it identical to the other
 - Do this using logical equivalences

Set Builder Notation and Logical Equivalences (Example)

- Show $\overline{A \cap B} = \bar{A} \cup \bar{B}$

Solution:

$$\overline{A \cap B} = \{x | x \notin A \cap B\}$$

Definition of Complement

$$= \{x | \neg(x \in A \cap B)\}$$

Definition of does not belong symbol

$$= \{x | \neg((x \in A) \wedge (x \in B))\}$$

Definition of intersection

$$= \{x | \neg(x \in A) \vee \neg(x \in B)\}$$

DeMorgan's Law

$$= \{x | (x \notin A) \vee (x \notin B)\}$$

Definition of does not belong symbol

$$= \{x | (x \in \bar{A}) \vee (x \in \bar{B})\}$$

Definition of Complement

$$= \{x | x \in \bar{A} \cup \bar{B}\}$$

Definition of Union

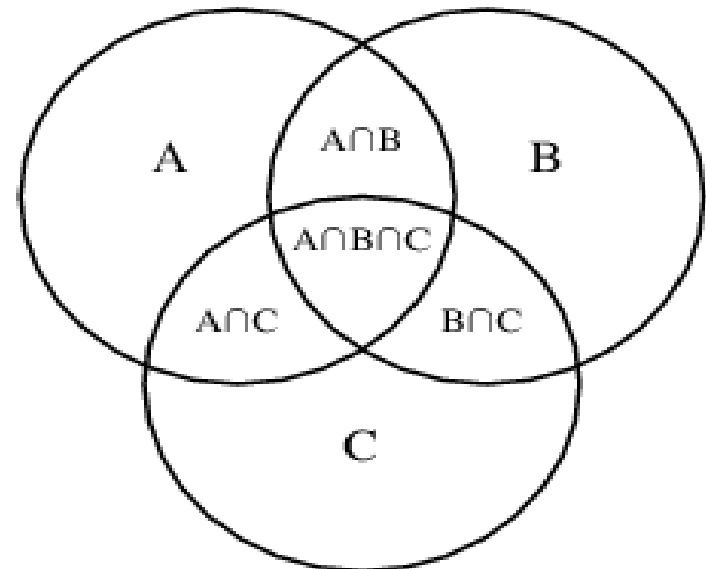
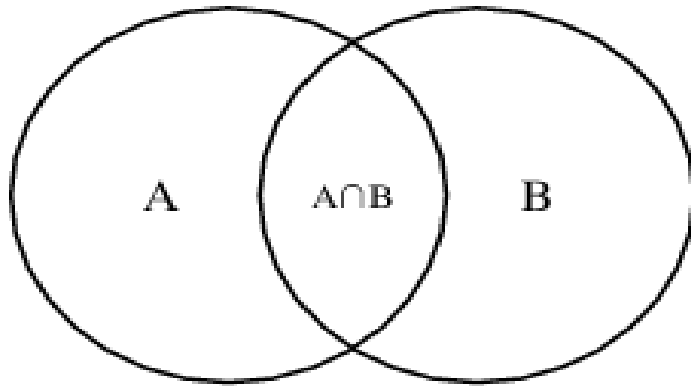
$$= \bar{A} \cup \bar{B}$$

By meaning of Set Builder Notation

2 and 3 - Set Venn Diagram

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



Example

Suppose a list A contains the 30 students in a mathematics class, and a list B contains the 35 students in an English class, and suppose there are 20 names on both lists.

Find the number of students:

- only on list A
- only on list B
- on list A or B (or both),
- on exactly one list.

Example

Solution:

- $30 - 20 = 10$ names are only on list A .
- $35 - 20 = 15$ are only on list B .
- $|A \cup B| = |A| + |B| - |A \cap B| = 30 + 35 - 20 = 45$.
- $10 + 15 = 25$ names are only on one list; that is,
 $|A \oplus B| = 25$.

Example

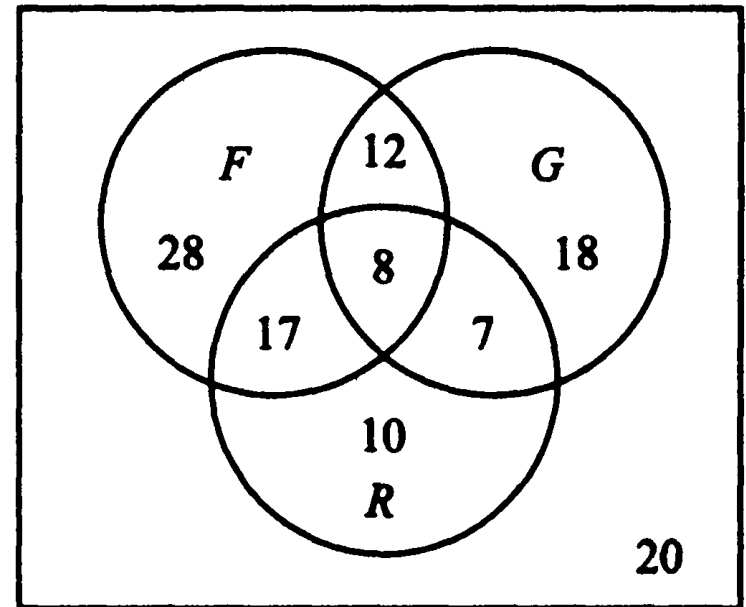
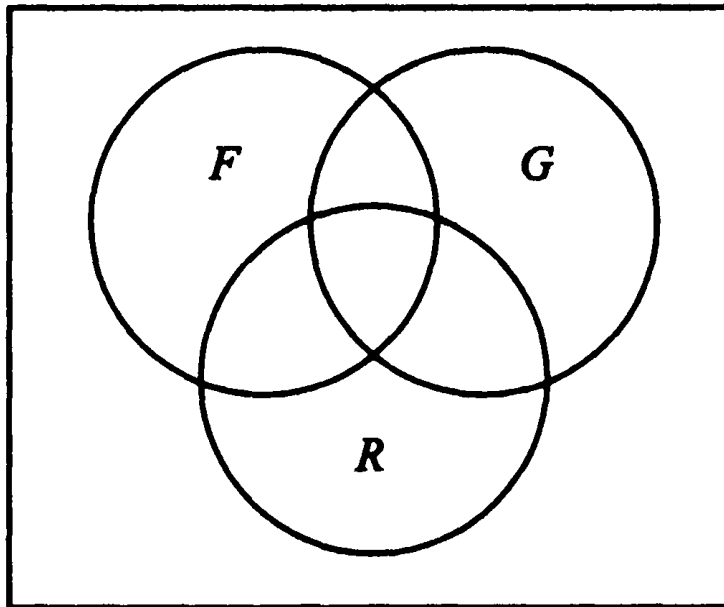
Consider the following data for 120 mathematics students at a college concerning the languages French, German, and Russian:

65 study French, 45 study German,
42 study Russian , 20 study French and German,
25 study French and Russian,
15 study German and Russian.
8 study all three languages.

Determine how many students study exactly 1 subject and fill the correct numbers of students in each eight region of Venn diagram shown in figure.

Example

- Total number of students exactly registered in one course
= $28+18+10=56$



Exercise Questions

Chapter # 2

Topic # 2.2

Question # 1, 2, 3,4,5,6,15,16,17,18,
19,20,21,22,23,24,25